

**Cost Optimisation Model
for Industrial Technologies (COMIT)**

Documentation and Technical Guide

[for external use]

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Please note: this guide provides an overview of the COMIT model, including its structure, data, and underlying methodology. It is not intended to be a comprehensive technical reference.

While care has been taken in preparing this material, users should exercise their own judgement when interpreting the information presented and, where appropriate, consult additional sources or expert advice.

The model, its documentation, and associated materials are subject to ongoing development and may be updated over time.

Introduction

The Cost Optimisation Model for Industrial Technologies (COMIT) is an optimisation model for UK industry, designed to explore when, where, and how Net Zero can be achieved by 2050.

COMIT is a site-level linear programming model that represents nearly 1,000 industrial sites across 17 sectors, each with distinct processes and over 400 technology options for deployment and decarbonisation. The model identifies the least-cost pathway for industrial decarbonisation, integrating hydrogen (H₂) and carbon dioxide (CO₂) infrastructure under various scenarios, constraints, or emissions targets. It includes key measures for decarbonising industry, including switching to low-carbon fuels, increasing energy and resource efficiency, and using carbon capture and storage.

The purpose of this document is to provide an overview of the COMIT model, including its data, assumptions and the mathematical formulation of the model's linear program (LP).

Background

Decarbonising industry presents significant challenges due to the diverse industrial sectors and technologies involved, scattered industrial sites, and the significant amount of infrastructure required for fuel supply and carbon capture. These complexities make it essential to adopt a systematic approach that couples technology deployment decisions with coordinated infrastructure planning. COMIT addresses these challenges by providing a tool for producing and assessing technically feasible, socially cost-effective pathways for industrial decarbonisation.

As modelled in COMIT, the industrial sector is driven by the need to meet industrial demand (for example, the amount of iron and steel that must be produced) over time at the least possible cost. To meet this demand, industrial sites build and use industrial 'technologies' (e.g. cement plants or iron and steel plants). Each industrial site in the model needs to invest in the right amount of technological capacity, which will allow it to produce the exact amount of output (e.g. Mt of cement) required. In each year, the model must make decisions about the amount of new technological capacity to use and build at each site, and how much hydrogen/CCS infrastructure to build. The aim is to meet the required industrial demand (a system constraint) while minimising the costs incurred, for a given set of assumptions and constraints around the cost and availability of different technologies (e.g. heat pumps, CCS), fuels, emissions, carbon values, and hydrogen/CCS infrastructure.

The COMIT linear programming model

Mathematically, COMIT is a linear programming model. Linear programming (LP), or linear optimisation, is a method to achieve the best outcome (such as maximum profit or lowest cost) in a mathematical model whose requirements and objective are represented by linear relationships. A linear optimisation problem formulation consists of three types of entities:

1. The decision variables: These represent the *choices* to be made by the model, i.e. the *unknowns*, or endogenous quantities, to be determined by the optimisation.

In COMIT, the decision variables are the amounts of individual technologies (e.g. heat pumps) and infrastructure (e.g. H₂ pipes) to be built or deployed at each site or cluster in each year. For example, the amount of new electric arc furnace capacity to invest in at the Port Talbot steel works in 2040, the amount of CO₂ to transport from Humberside to the North Sea storage site in 2025, and how much cement mixer technology to use at the Cemex Peterhead cement site in 2030.

2. The objective function: expressing the criterion to be minimised or maximised as a linear function of the decision variables.

For COMIT, the objective is to minimise total system cost. The objective function is the sum of eight different types of costs associated with the decision variables:

1. **Technology capital expenditure:** the cost of buying a unit of new technological capacity
2. **Fixed operating cost:** the cost of having a unit of technological capacity available on site
3. **Fuel:** the cost of the fuel needed to use a unit of technological capacity
4. **Carbon cost:** the amount of emissions released when using a unit of technological capacity multiplied by a carbon cost value
5. **H2 transport:** the combined capital expenditure and operating cost involved in transporting a unit of hydrogen built between a cluster and a site
6. **CO2 transport:** the combined capital expenditure and operating cost involved in transporting a unit of CO2 between a cluster and a site
7. **CO2 national transport:** the cost of shipping a unit of CO2 from a cluster centre to a storage site
8. **H2 national transport:** the combined capital expenditure and operating cost of transporting a unit of H2 between two clusters

3. The linear constraints: equations or inequalities involving the decision variables that restrict the values they can take. For example, a maximum amount of emissions, or the requirement for sites to produce enough output to meet a certain level of industrial demand.

In COMIT, there are two types of constraints: (i) the *essential* constraints that govern the model, and (ii) the additional ‘*user*’ constraints that can be specified by the user to explore different scenarios, such as constraining the amount of fuel that can be used by a particular technology.

The essential constraints express the physical and logical relationships that must be satisfied in order to represent the relevant industrial sector properly.

Examples of essential constraints include:

- the deployment of technologies must produce enough of each intermediate and final output commodity in each sector production pathway to meet required annual industrial demand.
- the amount of technological capacity available at each site is determined by the amount of existing capacity in the start year plus the amount of new capacity built in each year.
- the amount of hydrogen used by a site must be equal to the amount that has been transported to the site from a hydrogen production centre.
- the amount of new CO2 stored at a storage site each year must not exceed specified ‘injection’ limits.

Examples of user constraints include:

- limits on the amount of specific fuels that can be used in each year (e.g. biomass/hydrogen)
- limits on the amount of industrial emissions produced each year
- limits on the rate at which new technologies such as CCS and heat pumps can be rolled out across the UK

More details on the *user* constraints and how to use them are provided in the model user guide.

A complete formulation of the linear programming problem is provided in Appendix B. This LP formulation provides a good overview of how the optimisation model works and what assumptions and data contribute to COMIT’s decision-making process.

Industry sectors in COMIT

Background

In COMIT, the industries in scope are categorised into 16 industrial sectors plus refineries. Each sector has distinct processes, temperature requirements, and decarbonisation options that benefit from being modelled separately. Modelling industrial sectors separately allows for a more accurate representation of potential decarbonisation pathways. The list of industry sectors and their SIC (Standard Industrial Classification) code definitions is included in Appendix A.

Hydrogen production is usually excluded from the model (this is the default setup for the model), but the model can also endogenously model hydrogen production if required.

Industrial demand

In COMIT, 'demand' is the annual national industrial demand for the outputs produced by each sector, which must be met. This demand is either:

- (i) the final output produced by the sector in million tonnes (Mt), e.g. steel or cement
- (ii) the useful energy required in the sector in PJ, e.g. the amount of heat, refrigeration, motor energy, etc. required in the food and drink sector

As some sectors have multiple final output commodities, there are 21 output commodities contained in the **Demand_drivers** tab in the Input workbook. These figures represent the projected growth in industrial 'demand' over time. These demand projections act as a constraint in the model (i.e. the choices COMIT makes about which technologies to deploy must produce enough output to meet these commodity demands).

Industrial processes and technologies

Each industrial sector in COMIT has defined commodity production pathways, which are the means through which input fuels/commodities get translated into the sector's final outputs (e.g. steel, cement, paper). For those sectors with multiple final outputs (paper, chemicals, other industries), there are separate commodity production routes for each of those outputs. **Error! Reference source not found.** shows an example of the production pathways for cement.

Each sector production pathway involves one or more industrial processes. Processes in COMIT are the industrial processes (e.g. procedures involving chemical, physical, electrical, or mechanical steps) involved in producing certain industrial outputs. For each process, the model can usually select from different production technology options—for example, a coal-based technology versus an electricity-based technology for the same process.

In COMIT, a production technology represents a specific item of machinery, plant, or production method that transforms some commodities into other commodities. Technologies are categorised as either an actual technology, a generic process technology, or not an actual technology.

- **An actual technology** refers to a real, site-level technology deployed in industrial settings (e.g., boilers, heat pumps).
- **A generic process technology** represents a combination of multiple site-level technologies that sectors use to meet fuel requirements and maintain energy balance within the model.
- A technology classified as '**not an actual technology**' refers to dummy technologies that facilitate energy transformations in the model (e.g. heat transfers from CHP plants). These technologies have zero cost/capacity and therefore do not require updating.

Sector modelling

Industrial sectors in COMIT are modelled using either a process- or energy-based approach. For sectors modelled in a process-orientated manner, the actual production steps are modelled. For the remaining sectors, there is not enough information to model their production steps in full. Instead, they are split into different energy service demand categories (high-temperature heat, low-temperature heat, drying, motor drive, refrigeration, other, and low-temperature space heat), and generic technologies that produce these types of energy are modelled.

Process-based sectors: iron and steel, cement, lime¹, paper (excluding printing), glass, parts of the chemicals sector

- the end-use output is a product (e.g. cement)
- the technologies in the model represent the actual processes required to produce this product
- detailed data is available for these processes

These sectors can be modelled more accurately because the processes are more clearly defined.

Energy-based sectors: food and drink, non-ferrous metals, other industries, ceramics, rest of chemicals, printing (sub-sector within paper), mechanical engineering, electrical engineering, vehicles, textiles, construction

- the end-use output is the amount of energy required in the sector (useful heat/energy in PJ)
- the industrial technologies in the model represent generic processes that consume fuels to produce different types of energy required in the sector
- these sectors cover very diverse processes, and detailed data is not available, so the model is a significant simplification
- ECUK provides data on fuel consumption (coal, gas, oil and electricity) under different generic industrial processes, which is used to determine the processes and technologies for each of these sectors

Cement production pathway example

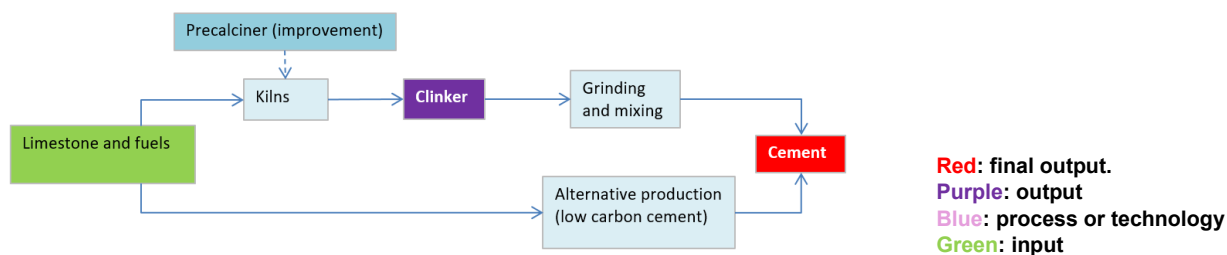


Figure 1 The cement industrial pathway, which is an example of a process-based sector (final and intermediate products are represented but not all input fuels are shown).

The cement production pathway is an example of a process-based sector (Figure 1). The output commodity for the cement sector is cement (in Mt). Cement is produced via grinding and mixing production technologies from limestone and clinker (and additional fuels). Clinker, an intermediate output, is produced by kiln production technologies from limestone, together with other input fuels such as electricity, natural gas and/or coal.

COMIT treats all sites within a sector as homogeneous, apart from size and location/access to infrastructure. This means that all cement sites model the full cement production process and have all cement technologies available to use.

¹ Lime is currently modelled based on cement processes, meaning the production steps are not completely accurate at present.

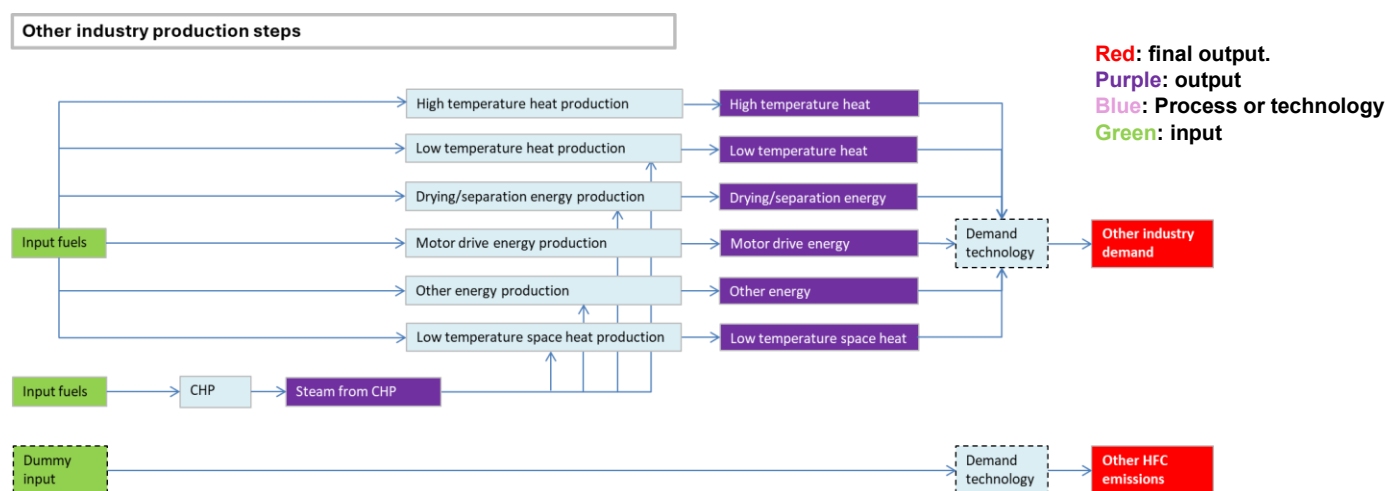
Table 1 List of production technologies in the cement sector, organised by process

Output commodity	Output commodity name	Process	Technology	Technology name
ICM	Cement (demand commodity)	Grinding and mixing	ICMGRIMIX01	Grinding and mixing technology
ICM			ICMGRIMIX02	Grinder and mixer with increased clinker substitution
ICM		Alternative cement production process	ICMLOCARB01	Alternative low carbon cement
ICMCLK	Clinker	Kiln	ICMKLNCLQ01	Calcium looping kiln
ICMCLK			ICMKLND01	Dry kiln, best available technology (BAT)
ICMCLK			ICMKLNMAQ02	Advanced Amine (MDEA) - BAT KILN (Cormos 2017) (CCS)
ICMCLK			ICMKLNMCO01	Dry kiln with coal CHP and MEA CCS
ICMCLK			ICMKLNMNQ01	Dry kiln with natural gas CHP and MEA CCS
ICMCLK			ICMKLNOXQ01	BAT kiln full oxyfuel (CCS)
ICMCLK			ICMKLNPOQ01	Partial oxyfuel dry kiln with CCS
ICMCLK			ICMKLNUST02	Fluidised bed kiln with waste utilisation

Table 1 shows the list of production technologies for the cement sector, organised by output and process. There are multiple technologies available for each cement process, which each use different fuels. The fuel inputs per unit of output commodity are captured in the **technology_input_output** tab. The core concept of COMIT is to enable the model to select among these alternative technologies for each stage of the production process, optimising deployment based on defined criteria.

Other industries production pathway example

For the sectors that are all or partly based on energy service demands, the 'output' from the production pathway is the amount of energy/heat supplied. The production pathway for the other industries sector is shown in Figure 2.

**Figure 2** The other industries industrial pathway, which is an example of an energy-based sector.

The processes in the other industries sector correspond to these industrial processes in ECUK, and ECUK data is used to calibrate the model so that modelled fuel use in COMIT in the starting year aligns with published energy statistics (i.e. ECUK/DUKES).

Figure 2 provides two examples of the dummy ('not an actual') technologies used in COMIT. They are represented by dashed outlines to indicate they are not real (for example, these technologies have no cost).

IOI01 is the dummy demand technology for the other industries sector. It takes as input the output commodities from the six types of processes represented in this sector (low-temperature heat, high temperature heat, etc.; the six purple outputs shown in Figure 2). It does not cost money to use (unlike standard production technologies) and provides a way for the model to represent the conversion of the outputs from the six generic processes into the required energy output for the sector. The inputs for this dummy technology are percentages set up to represent the relative share of each type of process required in that sector.

Another example is the artificial hydrogen_conversion sector and the three 'dummy' technologies it is made up of (INDMAINSHYGB_conversion, INDMAINSHYGG_conversion, INDMAINSHYGR_conversion). Unlike the dummy demand technologies above, these are created during the model run, and as such, are listed in the model outputs, but not in the Technologies tab in the Input template. More about these hydrogen_conversion sector technologies will be covered in the section on how hydrogen is handled in the model.

Table 2 List of production technologies in the other industries sector, organised by process

Output commodity	Output commodity name	Process	Technology	Technology name
INDHFCOTH	Dummy demand for other HFC emissions		INDHFCOTH01	Dummy demand for other HFC emissions
IOI	Demand commodity for the other industries (in PJ of useful energy)		IOI01	Demand technology (not an actual technology only transforms energy services (IFDLTH, IFDDRY, IFDMOT, IFDREF, IFDOTH) into final demand commodity (IFD))
IOIDRY	Drying / Separation services for the other industries	Drying/ separation energy	IOIDRYBIOS01	Drying / separation technology based on solid biomass
IOIDRY			IOIDRYELC01	Drying / separation technology based on electricity
IOIDRY			IOIDRYELCHP01	Drying / separation technology based on heat pump
IOIDRY			IOIDRYHCO01	Drying / separation technology based on coal
IOIDRY			IOIDRYHDG01	Drying / separation technology based on hydrogen
IOIDRY			IOIDRYLPG01	Drying / separation technology based on liquified petroleum gas
IOIDRY			IOIDRYNGA01	Drying / separation technology based on natural gas
			IOIDRYSTM01	Drying / separation technology based on low-temperature steam/heat from CHP (not an actual technology, only transfers heat from CHP plants, therefore no cost/capacity)
IOILTH	Low temperature heat for the other industries	Low temperature heat	IOILTHBIOS01	Low-temperature heat technology based on solid biomass
IOILTH			IOILTHELC01	Low-temperature heat technology based on electricity
IOILTH			IOILTHELCHP01	Low-temperature heat technology based on heat pump
IOILTH			IOILTHHCO01	Low-temperature heat technology based on coal
IOILTH			IOILTHHDG01	Low-temperature heat technology based on hydrogen
IOILTH			IOILTHLPG01	Low-temperature heat technology based on liquified petroleum gas
IOILTH			IOILTHNGA01	Low-temperature heat technology based on natural gas
IOILTH			IOILTHSTM01	Low-temperature heat technology based on low-temperature steam/heat from CHP (not an actual technology, only transfers heat from CHP plants, therefore no cost/capacity)
IOIMOT	Motor drive for the other industries	Electric motor	IOIMOTELC01	Electric motor technology

IOIOTH	Other energy services for the other industries	Generic	IOIOTHBIOS01	Generic process technology for other energy services based on solid biomass
IOIOTH			IOIOTHELC01	Generic process technology for other energy services based on electricity
IOIOTH			IOIOTHHCO01	Generic process technology for other energy services based on coal
IOIOTH			IOIOTHHDG01	Generic process technology for other energy services based on hydrogen
IOIOTH			IOIOTHLPG01	Generic process technology for other energy services based on liquified petroleum gas
IOIOTH			IOIOTHNGA01	Generic process technology for other energy services based on natural gas / biomethane
IOISPC	Low temperature space heat for the other industries	Low temperature space heat	IOIOTHSTM01	Generic process technology for other energy services based on low-temperature steam/heat from CHP (not an actual technology, only transfers heat from CHP plants, therefore no cost/capacity)
IOISPC			IOISPCBIOS01	Low-temperature space heat technology based on solid biomass
IOISPC			IOISPCELC01	Low-temperature space heat technology based on electricity
IOISPC			IOISPCELCHP01	Low-temperature space heat technology based on heat pump
IOISPC			IOISPCCHCO01	Low-temperature space heat technology based on coal
IOISPC			IOISPCHDG01	Low-temperature space heat technology based on hydrogen
IOISPC			IOISPCLPG01	Low-temperature space heat technology based on liquified petroleum gas
IOISPC			IOISPCNGA01	Low-temperature space heat technology based on natural gas
IOISPC			IOISPCSTM01	Low-temperature space heat technology based on low-temperature steam/heat from CHP (not an actual technology, only transfers heat from CHP plants, therefore no cost/capacity)
IOISTM	Low-temperature steam/heat from CHP for the other industries	CHP	IOICHPBIOS01	Steam turbine CHP based on solid biomass
IOISTM			IOICHPCCGTH01	CCGT CHP based on hydrogen
IOISTM			IOICHPGT01	Gas turbine CHP based on natural gas
IOISTM			IOICHPHP01	CHP based on heat pump and grid import

Industrial Technologies in COMIT

In COMIT, the industrial technologies represent all relevant production and decarbonisation options considered within the model. Each technology is represented in the model by three types of decision variables. These decision variables represent the amount of the technology newly built (called its *new capacity*), the total installed capacity (called its *available capacity*) and the amount used (called its *used capacity*) in each year. The linear optimisation model aims to solve for the optimal amounts of each technology to install and use at each site over time. This choice about what technologies to deploy, when, and by how much can be referred to as a 'deployment profile' (or alternatively, 'technology rollout' or 'pathway').

The input template workbook holds the input assumptions for the COMIT technologies, defined in the (i) **Technologies** and (ii) **technology_input_output** sheets.

The **Technologies** tab sets out the parameters defined for each technology included in the model, including lifetime, costs, efficiencies, first year of availability, available capacity at the model start year, and units of output/capacity (see **Table 5** for a full list of parameters).

The **technology_input_output** sheet shows, for each technology, what commodities are input and output per unit of its main output (i.e. its fuel use profile). For example, a natural gas boiler in the paper sector (IPPBOINGA01) uses 1.11 PJ of natural gas (IND_NGABOM) for every PJ of low-

temperature heat (IPPLTH) produced. In the **technology_input_output** tab, this relationship would be represented with a row for the input and a row for the output, as in Table 3.

Table 3 Example of a possible input/output relationship for IPPBOINGA01 in the technology_input_output worksheet. There is a separate row for each input and output. Inputs are negative, and amounts are standardised to show commodities in/out per unit of main output.

<i>technology_code</i>	<i>commodity_label</i>	<i>amount of input/output</i>
IPPBOINGA01	IND_NGABOM	-1.11
IPPBOINGA01	IPPLTH	1

All relevant inputs to technologies and outputs from technologies, including process emissions outputs, are called “commodities” and must be listed in the **commodities** tab of the Input workbook.

Table 4 The technology assumptions in COMIT

COMIT Technology parameter	Description
<i>Technologies tab</i>	
Code	Unique identifier for the technology
Name	Description of the technology
Sector	Industrial sector
output_commodity	The output commodity produced by the technology
emissions_released	Emissions released per unit of commodity produced
existing_capacity_2021	The amount of existing (i.e. already built) capacity of the technology at the model start year (2021).
capacity_to_activity_factor	Capacity to activity conversion factor i.e. amount of capacity purchased for the cost listed under NCAP_COST
availability_factor	Availability factor: maximum annual amount of main output commodity per unit of capacity
Capex	Capital costs per unit of new capacity
fixed_opex	Annual fixed non-fuel O&M (operating and maintenance) costs per unit of capacity
Lifetime	Lifetime of technology (i.e. any new capacity built)
start_year	Year in which technology is first expected to be available
technology_category	Technology grouping according to fuel type used (e.g. coal, oil)
output_unit	Unit in which main output commodity is measured
retrofit_to	
<i>technology_input_output tab</i>	
technology_code	Unique identifier for the technology
Commodity	Unique identifier for the commodity
Output	Commodity in/out per unit of main output
primary_commodity	Primary commodities are inputs that are not outputs from another technology
commodity_produces_emissions	Apply emissions factor (Y = apply emissions factor from commodity table, N = no emissions from input commodity)

Baseline technological capacities

The initial or baseline capacity for each technology in COMIT represents the amount of built capacity assumed to exist at the model start year (see existing_capacity_2021 in Table 4). Because detailed data on actual installed capacity by technology in 2021 is unavailable, these baseline capacities are estimated to align COMIT’s modelled fuel use with published energy statistics (e.g., DUKES, ECUK).

It is assumed that there is just enough capacity in the start year to meet industrial demand, with no spare capacity.

Fuel use 'profiles' for each technology are the inputs per unit of main output specified in the technology_input_output tab. These are adjusted by the availability_factor² and then multiplied by the existing capacity amount for a given technology to calculate starting year fuel use.

For each process, the output from the corresponding fuel/technologies for that process is:

Implied output = existing_capacity_2021 * capacity_to_activity_factor * availability_factor

These starting technology capacities are then apportioned to individual sites based on the proportion of NAEI emissions for that sector they represent (in a similar manner to site demand).

Interpreting technology codes

Each sector has been allocated a specific code (for example, the code for cement is 'ICM'). In most cases, this also corresponds to the code for the output (demand) commodity for that sector. The exceptions to this are subsectors such as high-value chemicals and printing, which have their own demand commodities. The demand commodities used in COMIT can be found in the Demand_drivers tab in the Input workbook.

For technologies, the first three digits of the technology codes correspond to their sector code. For the majority of technologies, the next three digits then indicate the industrial process involved, and the three digits after that indicate the type of fuel being used. For example, IOIDRY indicates technologies for drying in the Other industries sector (IOI), and IOIDRYELC01 uses electricity (ELC) while IOIDRYNGA01 uses natural gas (NGA).

For sector-specific technologies—such as those in cement—the codes won't necessarily align exactly with the 3-3-3 pattern described above, but they can be interpreted in a similar way. In cement, for example, all kilns are labelled with KLN, and the later digits of the technology code will usually indicate the main fuel type used by the technology (e.g. BIO is usually in the code for technologies that use biomass). Table 5 provides a reference for some of the key sector, process and fuel codes in COMIT.

Codes are also used in the model to refer to the various commodities being used or produced. These codes often use the same or similar abbreviations as the technology codes. In Table 1, for example, the cement-specific commodity clinker (ICMCLK) starts with the abbreviation for the cement sector (ICM), followed by an abbreviation of 'clinker' (CLK). The outputs from the refrigeration technologies in the model (IFDREF, IOIREF, ICHREF) start with the sector code and end with 'REF', the abbreviation/code for refrigeration.

Table 5 List of the technology/process codes in COMIT

Sector / subsector	Code	Process	Process codes	Other codes	
<i>Ammonia*</i>	IAM	High temperature heat	HTH	HP	heat pump
<i>Chemicals</i>	ICH	Low temperature heat	LTH	SC	Steam cracker
<i>Cement</i>	ICM	Drying	DRY	KLM	Kiln
<i>Food&Drink</i>	IFD	Motoring	MOT	BIO	Boiler
<i>High value chemicals*</i>	IHVC	Refrigeration	REF	FUR	Furnace
<i>Iron&Steel</i>	IIS	Other	OTH	CHP	CHP technologies
<i>Non-ferrous metals</i>	INF	Low temp space heating	SPC	COG	Coke oven gas

² Availability factors are the maximum annual amount of main output commodity per unit of production capacity. Over the course of a year, a production technology may not be operating at full capacity the whole time, e.g. due to maintenance periods. For COMIT technologies, availability factors are typically about 60-90%.

<i>Ceramics&Other</i>	ICR		
<i>Other</i>	IOI		
<i>Paper production</i>	IPP	Fuel	Fuel codes
<i>Refineries</i>	PRCOIL	Natural gas	NGA
<i>Glass</i>	IGL	Coal	HCO
<i>Mechanical engineering</i>	IME	Biomass	BIO
<i>Electrical engineering</i>	IEE	Liquified petroleum gas	LPG
<i>Vehicles</i>	IVH	Hydrogen	HDG
<i>Textiles</i>	ITX	Electricity	ELC
<i>Construction</i>	ICN	Low-temperature steam/heat from CHP	STM
<i>Printing*</i>	IPR		
<i>Lime</i>	ILM		
<i>Hydrogen</i>	PHYG		

**Indicates a subsector*

Treatment of resource efficiency and energy efficiency (REEE)

In COMIT, resource efficiency (RE) and energy efficiency (EE) measures are modelled exogenously. Their impacts are treated as a proportional reduction in demand for final outputs. To implement REEE measures, the model takes as input the resource and energy efficiency measures (in the **resource_efficiency** and **energy_efficiency** tabs in the Input workbook) and these are then applied as a percentage reduction in the national demand for the output commodity for each sector (see **Demand_drivers** sheet):

$$\text{Adjusted_demand} = \text{demand} * (1 - \text{efficiency})$$

For example, a 2% resource efficiency measure in 2021 for Ammonia (code: IAM), would mean:

$$\text{Adjusted_demand} = 0.7269 * (1 - 0.02) = 0.712 \text{ Mt}$$

These demand adjustments are calculated at the beginning of the model setup, and the model then solves to meet this adjusted demand.

Resource efficiency measures are applied before energy efficiency measures.

$$\text{Adjusted_demand} = \text{demand} * (1 - r_efficiency) * (1 - \text{efficiency})$$

Site level modelling

The industrial sites in the model are based on NAEI (National Atmospheric Emissions Inventory) data from 2021. This dataset contains site emissions data for industrial and commercial 'point sources' in the UK, i.e. emission sources at known locations. The industrial sites in the NAEI dataset are mostly point sources registered with, and trading emission credits under, the UK Emissions Trading Scheme (UK ETS). However, it also contains some non-traded emissions, reported via pollution inventories, and a small number of 'modelled' emissions. Approximately 10% of the NAEI is non-traded.

The NAEI dataset contains point source names, sectors, emissions in tonnes of CO₂, and locations in easting and northing.

Every site is allocated to a cluster region based on its closest distance to a cluster centre point. Sites outside the cluster radius (typically 30 km) are referred to as 'dispersed sites'.

Point-source sites emitting less than 10 ktCO₂e within each region are aggregated to reduce the total number of sites and improve computational efficiency. These aggregated sites are assigned a representative location based on their average distance from the cluster centre. In the code and output file, these grouped small sites can be identified by the suffix '_psg' (i.e. point-source-grouped).

Non-point source sites

Non-point sources are those that are not covered in the NAEI dataset and for which we have very limited data. Non-traded sites are a mixture of point sources and non-point sources. Emissions not covered by NAEI data are treated as "non-point" sources.

To account for non-traded emissions from strictly non-NAEI sites, we create artificial non-point sites. Two artificial non-point-source sites are created for every cluster and sector: one inside the cluster radius and one outside the cluster radius. Each of these artificial modelled sites represents multiple sites in the real world. In the code and output file, these grouped small sites can be identified by the suffix '_npsg' (i.e. non-point-source-grouped).

They are assigned an average distance from the cluster centre and latitude/longitude based on point-source sites in that cluster/sector that are inside or outside the cluster radius, respectively.

Creating two artificial sites per cluster/sector combination gives us more control over whether to allow non-point-source sites to connect to clustered infrastructure if they are close enough.

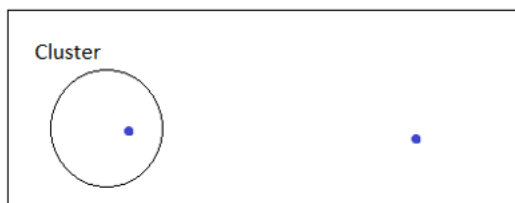


Figure 3 Representation of the two artificial non-point-source sites created for each cluster in COMIT, to represent clustered and dispersed sites.

Since many of the sites in COMIT are artificial, care should be taken to avoid reporting these as ‘real’ sites. Generally, COMIT is designed to reflect average site behaviour rather than provide specific site-level detail.

Site demand

For each sector, total industry demand is allocated among all sites for that sector. At the site level, the outputs from the technological processes at each site must then sum to meet this site-level demand for each commodity at each timestep. Site-level demand therefore changes each year in line with the sector-level demand projections.

The model splits site demand into three categories: traded, non-traded point sources and non-traded non-point sources (Figure 4):

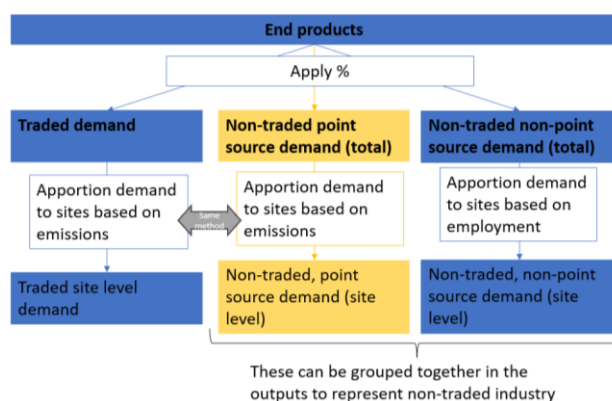


Figure 4 Schematic illustrating the apportioning of demand amongst traded point-source and non-traded point-source and non-point-source sites.

For estimating site-level demand:

1. The model starts with total demand for each sector and applies a traded/non-traded percentage to get traded demand and non-traded demand by sector.
2. The model splits traded demand among traded point sources according to their emissions.
3. The model splits non-traded demand among non-traded point sources and non-point sources.

The size of the demand to apportion to traded and non-traded sites is determined by traded/non-traded splits (as can be found in the data template – see **traded_share** tab).

Traded point-sources

For traded point-source sites, the site's annual emissions are used as a proxy for site size and to weight the proportion of demand allocated to each site:

Site demand for sector X = [ratio of NAEI site emissions to total traded emissions for the sector] * [total traded sector demand]

Non-traded point-sources

The non-traded portion of demand is split between non-traded point sources and non-point-source sites. As with traded point sources, for non-traded point-source sites, the site's annual emissions are used as a proxy for site size and to weight the proportion of demand allocated to each site.

Site demand for sector X = [ratio of site emissions to total non-traded point-source emissions for the sector] * [total non-traded point-source sector demand]

Non-point sources

Non-point sources are all non-traded. Demand is split among non-point sources by weighting them according to their employment numbers using IDBR data. Employee numbers are used as a proxy for site size.

Industrial clusters

Industrial clusters are geographic areas with a high concentration of industrial activity. In the UK, nearly half of all industrial emissions originate from these clusters, while the remainder comes from “dispersed sites”, defined as sites located outside cluster boundaries.

COMIT defines ten cluster regions across Great Britain. Each cluster is represented by a central point and includes all sites within a specified radius (λ), typically 30 km. Sites within this radius are assumed to have access to the cluster's hydrogen (H₂) and carbon dioxide (CO₂) infrastructure. Sites outside the radius (the ‘dispersed sites’) are assumed not to be able to access this infrastructure.

Cluster locations are specified in the **Cluster_location** tab using latitude and longitude coordinates. Sites are assigned to clusters based on the shortest distance to the cluster centre.

The model is designed to be flexible and allows for adjustments to be made to cluster definitions. However, changes should be made carefully, as they affect multiple components in the model.

Options for changing cluster definitions include:

- deactivating existing clusters by setting use_cluster = FALSE in the **Cluster_locations** tab.
- modifying the cluster radius for each cluster in the **cluster_radius** tab.
- adding new clusters, or revising existing ones, by changing the list of clusters and their latitude and longitude in the **Cluster_locations** tab.

Infrastructure for hydrogen and CCS

In COMIT, hydrogen and CCS infrastructure are assumed to be developed at cluster centre points, in line with UK policy that prioritises clusters for the initial rollout of CCS and hydrogen networks. Concentrating infrastructure in clusters, where multiple high-emitting facilities are located close together, enables shared use of assets, reducing costs and accelerating implementation.

Hydrogen infrastructure

In COMIT, hydrogen can be transported from a hydrogen production plant to a site using distribution technologies such as pipelines, trucks, shipping, *etc.* We assume hydrogen production plants are built at the centre point of each cluster (hydrogen point). For each cluster, there is a specified first year in

which infrastructure for hydrogen production and transport becomes available. These first possible years of deployment for hydrogen are defined for each cluster in the **cluster_radius** tab (in the **H2_first_year** column).

Starting in these years, sites within the cluster radius then have the option of accessing hydrogen from the nearest hydrogen point using one of up to three transport options:

1. **New-build direct pipeline:** this involves building a **new** pipeline from the H2 production source to the industrial site, supplying that site only.
2. **Trailer hydrogen:** hydrogen is compressed and transported in a tube trailer to the industrial site (referred to as 'trucking' in the rest of this section and in the model).
3. **Grid hydrogen:** in the future, hydrogen may be delivered to industrial sites via a converted gas grid network, resembling a 'hydrogen grid'. Sites will pay a network cost, much like under the current system. The network cost **does not** vary with distance from the production source, making it different from the other two transport options.

Hydrogen transport costs

The cost of transporting hydrogen to a site depends on three factors: transport method, quantity of hydrogen, and distance to the hydrogen source. Hydrogen transportation costs by distance and capacity for the model are specified in the **H2_transport_cost** tab.

For pipeline construction, similar to new production technologies, sites incur both capital expenditure (capex) and operating expenditure (opex). COMIT allows pipelines to be built incrementally rather than all at once. Capex and opex (in £m) vary by distance (km) and capacity (MW), allowing the model to capture economies of scale.

Pipelines can only be built in discrete capacity sizes. For each site, the smallest pipe that meets its maximum expected hydrogen demand is selected.

Cost data is provided for each possible distance and capacity size, meaning that hydrogen transport costs are calculated by matching (i) the site's distance to the hydrogen production location, and (ii) the maximum potential hydrogen expected to be used, to the available distance and capacity categories.

For site distance, the model selects the data point above the site's actual distance. The flow rate data point selected is the one less than the actual flow rate. The **sector_max_H2_CO2** tab gives the maximum potential PJ of hydrogen for each unit of output produced. This is multiplied by the site demand to give the max_H2 requirement (i.e. flow rate) for that site.

Within the model, for each flow rate, a linear model is fitted to determine the capex and opex based on the site's exact distance to the hydrogen point.

The costs of transporting hydrogen by trucking and gridlines are calculated in the same way as pipeline costs, but with only an annual operating cost incurred and no initial upfront cost. For gridlines, the rate varies with flow rate only, not distance.

Specifying the hydrogen transport option

The **H2_transport_cost** tab allows the user to specify a preferred method for transporting hydrogen for each distance-from-cluster and capacity combination.

For each combination, the variable **Lowest_cost_option** can be set to 1, 2, or 3 according to whether trucking, pipelines, or gridlines represent the preferred (e.g. cheapest) option, respectively. Based on whether trucking, pipelines, or gridlines are selected as the **Lowest_cost_option**, the corresponding opex and/or capex cost information from either the grid, trucking, or pipe columns will be used to determine transportation costs for sites matching that distance and capacity combination (only pipelines incur capex costs). At present, only one transportation method is allowed per site.

For example, if the **Lowest_cost_option** is set to 2 for all combinations, that means only pipelines will be used to transport hydrogen.

Grid hydrogen

The `use_H2_grid_site` parameter in the **model_parameters_b** tab enables COMIT to model hydrogen delivery to industrial sites via a converted gas grid network. Gas grid conversion involves decommissioning parts of the existing gas network and repurposing them for hydrogen transport.

When `use_H2_grid_site` = TRUE, the **H2_grid** tab specifies the first year each NAEI (point-source) site can access hydrogen through a national grid connection. For non-point-source sites, if `use_nps_for_H2_grid_sites` is set to TRUE on the **model_parameters_b** tab, non-point-source sites will be able to access the grid at the same time as nearby point-source sites; if it is set to FALSE, they will not be able to access the grid.

CO2 storage

Carbon capture and storage (CCS) involves capturing the CO₂ produced at industrial plants or power stations; transporting it by pipeline, road and/or ship; and then permanently storing it deep underground, for example in offshore storage sites.

Five clusters (terminals) have direct access to offshore CO₂ pipeline transport infrastructure and a designated storage site to which CO₂ is shipped (Table 6). Other clusters transport CO₂ to one of these five terminals.

The **CO2_T&S_cost** tab defines costs (£m/kt) for the transport and storage of CO₂ from clusters to storage sites for all possible shipping, transport, and storage routes. This cost is a fixed price and covers transport (shipping) from cluster point to terminal and storage, based on the average amount of transport and storage involved.

There is a constraint on the maximum amount of CO₂ that can be stored per year at each of the four storage sites. These are defined as maximum annual injection rates (kt/y) in the **CO2_storage** tab.

Table 6 The cluster terminals and their corresponding storage region

Terminal(s)	Accessible Storage Region
Peterhead	Northern & Central North Sea
Teesside, Humberside	Southern North Sea
Merseyside	East Irish Sea
Southampton	Southampton Sea

For each cluster, as with hydrogen, there is a specified first year in which infrastructure for CO₂ transport and the storage point become available. These first possible years of deployment for CCS are defined for each cluster in the **cluster_radius** tab (in the `CO2_first_year` column).

Starting in these years, sites have the option to develop dedicated pipelines or use trucking to transport CO₂ to their nearest terminal or cluster point locations (i.e. with the additional capex/opex borne by sites in the model, along with the levelised £/tCO₂ T&S costs at each terminal or cluster point).

CO2 transport costs

The cost of CO₂ transportation for a site is calculated in the same way as for hydrogen, but with options only for trucking or pipelines (no grid). For pipelines, sites incur both an initial construction cost and an ongoing operating cost. For transporting CO₂ by trucking, there is only an operating cost and no initial upfront cost.

The capex and opex (in £m) of CO₂ transportation depend on the distance (km) and capacity (MW) of the pipe, allowing the model to reflect economies of scale. For pipelines, the model assumes that they

can only be built in certain discrete capacities. For each site, the smallest pipe that will meet its maximum expected CO₂ capacity (flow rate) is chosen. Refer to the H₂ transport costs section for full details on how the pipeline capex/opex costs for each site are calculated.

For each distance/capacity combination, the variable **Lowest_cost_option** in the **CO₂_transport_cost** tab can be set to 1 or 2 according to whether trucking or pipelines represent the preferred option. The corresponding opex and/or capex cost information will then be used to determine transportation costs for sites matching that distance and capacity combination. At present, only one transportation option can be used by a site.

The Peak Spur cluster

This functionality adjusts CO₂ pipe distances for a specified set of Peak District cement and lime sites. It is used to account for the Peak Cluster spur, where infrastructure is planned to connect these sites to Merseyside's CCS. When enabled, the feature updates the pipeline distances for these sites to reflect the route to a nearby Peak Spur cluster point (see the **CCS_spur_sites** tab). This changes the CO₂ transport costs for these sites and also means they can access CCS even when access to CO₂ infrastructure is restricted to locations within the cluster radius.

The **Use_Peak_CCS_Spur** parameter on the **model_parameters** tab can be used to turn this functionality on or off.

Carbon values

In COMIT, a carbon cost is added to the objective function to penalise emissions. This is used to reflect the cost of emissions and incentivise decarbonisation. Without these carbon costs in the model, industry would not decarbonise.

In COMIT, the carbon cost is calculated as the amount of carbon emitted at each site multiplied by an annual carbon value, usually expressed as £ per tonne of CO₂e, with separate carbon values for traded and non-traded sites.

Carbon_cost = sum [traded_carbon_value * CO₂ emissions for traded sites + non-traded_carbon_value * CO₂ emissions for non-traded sites]

Emissions

In COMIT, total site emissions are categorised into two components: energy-related (fuel) emissions and process emissions.

- Fuel-related emissions arise from burning fuels for energy or using them as feedstock.
- Process emissions result from chemical or industrial reactions, such as calcination in cement production.

Both fuel and process emissions are considered direct emissions, as they occur at the point of use within the facility.

Emissions are reported in terms of carbon dioxide equivalent (CO₂e), which accounts for both CO₂ and non-CO₂ greenhouse gases (e.g. CH₄, N₂O).

Fuel emissions

Fuel-related emissions result from the combustion of fuels used to provide energy for industrial processes. Fuel emissions for each industrial technology are estimated by multiplying the technology's activity level, represented in COMIT as used capacity, by the relevant emissions factor. Emissions are expressed as net emissions per unit of used capacity in kilotonnes of CO₂e.

The emissions factors for fuels are provided in the **Fuel_emissions** tab and are defined as greenhouse gas emissions (ktCO₂e) per unit of fuel consumed (PJ). For some fuels, these factors vary over time.

To view fuel emissions in the output workbook, go to the **Emissions** tab and select 'Energy' under **Emissions_type**.

Process emissions

Process emissions are greenhouse gases released from chemical or physical reactions during industrial processes, rather than from fuel combustion for energy. In COMIT, they are calculated using technology-specific input/output data and emissions factors. The calculation is based on process emissions per unit of *used capacity*, adjusted for the technology's CO₂ capture rate. Process emissions are treated as outputs from technologies and are listed in the **technology_input_output** tab of the input workbook, expressed as emissions per unit of activity, where activity refers to the technology's main output. To identify commodities that generate process emissions, filter the **commodities** tab for `process_emission = TRUE`.

To view process emissions in the output workbook, go to the **Emissions** tab and select 'Process' under **Emissions_type**. The **DetailedTables** summary tab also has a section showing captured and process emissions for each sector, from row 44.

Traded emissions

Emissions can be classified as either traded or non-traded. Traded emissions come from sites covered by the UK Emissions Trading Scheme (ETS), whereas non-traded emissions do not fall within the scope of the UK ETS. Traded and non-traded emissions have different carbon prices associated with them (see the **Carbon_price** tab in the Input workbook).

The **traded_share** tab in the Input workbook provides the traded and non-traded split of CO₂ emissions for industrial sectors. The figures are at sector level and are not split by fuel type. Note that this only covers CO₂ emissions, not other greenhouse gases.

Captured emissions

Captured emissions are the CO₂ emissions captured by carbon capture and storage (CCS), which are then transported off-site and stored in underground storage sites.

To view summary-level information about captured emissions in the output workbook, go to the section beginning at row 44 in the **DetailedTables** tab. The full breakdown for captured emissions is listed in the **Emissions** tab. To view it, go to the **Emissions_category** column (column A) and select 'Captured'. The captured emissions for each timestep will equal the amount of CO₂ transported at each timestep.

CCS technologies can be viewed in the **Outputs** tab by selecting 'CCS' under the **Technology_category** column. Similarly, in the **Technologies** tab in the Input workbook, select 'CCS' under the **Technology_category** column. The **emissions_released** column in the **Technologies** tab shows the emissions released per unit of commodity and is used to store the total emissions attributed to each technology. The **emissions_released** parameter will be 100% for all standard technologies and less than 100% for technologies that capture emissions (i.e. CCS technologies).

Indirect emissions

COMIT accounts for the indirect emissions associated with energy commodities that are produced outside the industrial site and supplied to it, rather than generated on-site. These are the emissions associated with the production of electricity (generated in the power sector) and hydrogen (produced via various pathways, e.g. blue, green and grey). They are calculated by multiplying the site's consumption of electricity and hydrogen by their respective emissions factors.

They can be viewed in the output workbook under the **Emissions_total** tab by selecting 'Electricity' or 'Hydrogen'.

Negative emissions

Negative emissions occur when a process captures more CO₂ than it emits.

In COMIT, negative emissions are calculated as **emissions before capture minus captured emissions**, and appear as negative values in the emissions outputs. They are important for achieving Net Zero targets because they offset residual emissions from hard-to-abate sectors.

Negative emissions can be viewed in the output workbook under the **Emissions_total** tab, where rows labelled "Negative" show these values by sector and year.

Non-energy use

Non-energy use refers to the consumption of fuels or feedstocks for purposes other than energy generation. Common examples include the use of fossil fuels as chemical feedstocks, such as naphtha for olefin production and natural gas for ammonia synthesis.

In COMIT, non-energy use fuels are modelled as inputs that do not produce energy outputs. These commodities are listed in the **Commodities** tab and can be identified by filtering the **commodity-category** for *NonEnergyUse*.

In the output workbook, non-energy use appears in the **Energy** tab under the fuel category *NonEnergyUse*. When calculating total fuel consumption for the energy sector, non-energy use is excluded, so it often needs to be filtered out when compiling fuel use statistics.

Note: units shown for *NonEnergyUse* in the table are not accurate (TWh) and should be in other units; The (incorrect) TWh units are there only for convenience in generating the output table.

Hydrogen in the model

As outlined in the problem formulation section, COMIT includes the capability to explicitly model hydrogen production. In this 'hydrogen production mode', hydrogen is an internally produced commodity, generated at hydrogen plants located at the centre of each cluster using the available hydrogen technology options (see the hydrogen sector technologies in the Input data template).

However, enabling hydrogen production changes the model to a mixed integer linear program (a MILP, rather than just an LP), and significantly increases solution time. In most cases, COMIT is run with this functionality turned off via the `model_H2_production` parameter (set to false) in the **model_parameters_b** tab. When this parameter is off, the model assumes hydrogen is available as a ready commodity, with predefined costs and emissions factors.

In non-production mode, the model allows for three types of hydrogen commodities: blue, green and grid hydrogen. They each have different costs and emissions associated with them, as listed in the **Fuel_costs** and **Fuel_emissions** tabs. The model selects the optimal mix of these three hydrogen types to use, based on cost and other constraints.

To operationalise this, COMIT uses a dummy sector called **hydrogen_conversion**, which comprises a single dummy site and three hydrogen conversion technologies (one for each hydrogen type). Hydrogen-consuming technologies use a generic commodity, **INDMAINSHYG**, which has no cost. Actual costs come from the specific hydrogen types (blue, green and grey), which are converted into a generic output commodity called **HYGEN** via hydrogen conversion technologies.

A **hydrogen conversion constraint** is used to ensure that the total hydrogen consumed by technologies equals the amount produced in the hydrogen conversion sector, totalled across the different hydrogen types. It forces the model to balance hydrogen consumption with production from costed sources.

To see the amounts of the different hydrogen commodities used in the model outputs, look at the use of the different hydrogen technologies in the **Outputs** tab. In the 'Sector' column, select 'hydrogen_conversion'. This will show the three dummy hydrogen_conversion technologies (INDMAINSHYGB_conversion for blue hydrogen, INDMAINSHYGG_conversion for green hydrogen, etc.) and their usage over time. Equivalent information for the hydrogen_conversion dummy sector technologies can be found in the **Costs** and **Energy** tabs.

By default, the model will usually select the cheapest hydrogen source—typically blue hydrogen—unless additional constraints are applied. To manage this, a common approach is to add a constraint on the *Maximum Capacity* tab that limits the share of blue versus green hydrogen. This constraint is expressed as a percentage relative to total hydrogen production. To implement this, a limit is placed on the **INDMAINSHYGB_conversion** technology (representing blue hydrogen). The percentage entered in this column caps the maximum contribution of blue hydrogen to overall hydrogen availability. For example, if the limit is set to 65%, the model can allocate up to 65% of hydrogen production to blue hydrogen. Beyond that threshold, it must switch to green hydrogen, which comes at a higher cost.

Combined Heat and Power (CHP)

In COMIT, combined heat and power (CHP) systems are modelled as technologies that produce both electricity and useful heat from the same fuel input. They output low-temperature steam or heat for industrial processes and produce electricity, which is assumed to be sold.

Fuel consumption and associated emissions are calculated based on the total fuel input, with outputs allocated between electricity and heat according to technology-specific efficiency parameters. Each CHP technology in COMIT has defined electrical and thermal efficiency values. These determine how much fuel is required for a given output mix. For example, a gas-fired CHP might allocate 30–40% of fuel energy to electricity and the rest to heat.

CHP technologies are identifiable by 'CHP' in their technology code and description. **Error!**

Reference source not found. shows the industrial production pathways for the Textiles sector, with the CHP pathway highlighted.

To represent the conversion of steam output from CHP into usable energy for the sector, COMIT employs 'dummy' technologies. These are not real technologies but serve as placeholders for heat transfer. They are marked with 'STM' in their technology code and labelled as "Low-temperature heat technology based on low-temperature steam/heat from CHP (not an actual technology, only transfers heat from CHP plants)."

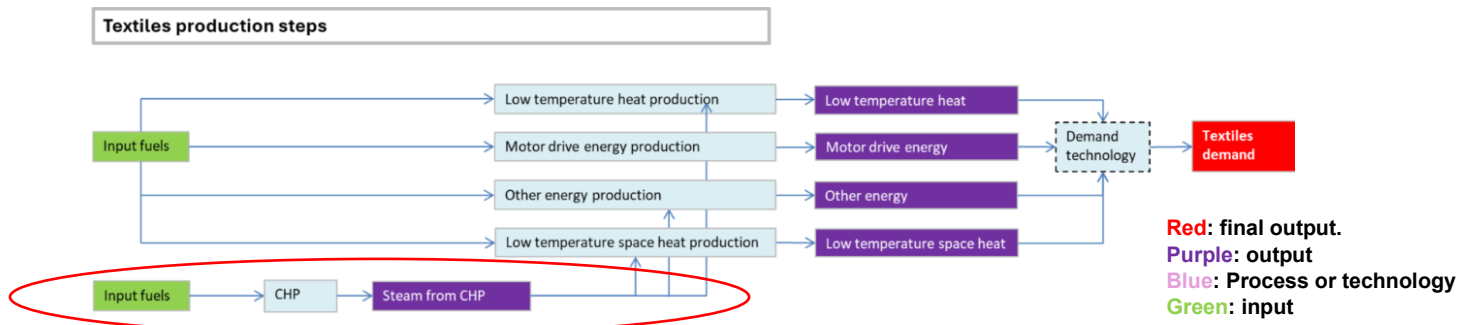


Figure 5 Industrial production pathways for the Textiles sector. The CHP pathway is highlighted.

Appendix A - Sector definitions

Sector	SIC Codes	Notes
Cement	23.51	
Lime	23.52	
Chemicals	20-21	Including pharmaceuticals (21). High value chemicals (IHVC) is within 20 (20.14)
Food&Drink	10,11,12	
Iron & Steel	24.1-24.3	Not including coke (19.1). Including blast furnaces row from DUKES (no specific SIC code for blast furnaces)
Non-ferrous metals	24.4-5 (excluding 24.46)	Excluding nuclear fuel (24.46)
Ceramics	8, 23 (excluding 23.5 and 23.1)	Includes other non-metallic minerals (rest of 23) and other mining and quarrying (8)
Other Industry	16, 22, 31-33, 36-39	Including water, sewage and waste (36-39)
Paper	17-18	Includes printing and publishing (18)
Refineries	19.2	
Glass	23.1	SIC code definition added
Mechanical engineering	25, 28	
Electrical engineering	26, 27	
Vehicles	29-30	
Textiles	13-15	
Construction	41-43	

Appendix B - Complete LP problem formulation

Decision variables

The objective function is a function of the decision variables below. These are the variables that the solver changes to find the extremum of the objective function:

- $\text{new_capacity}(t,s,\text{tech})$ – capacity of industrial technology tech , built in period t at site s .
- $\text{available_capacity}(t,s,\text{tech})$ – available capacity of technology tech , in period t at site s . This refers to capacity that is readily available at a site because it has either been newly built or built in a previous period and still exists.
- $\text{used_capacity}(t,s,\text{tech})$ – actively used capacity of technology tech , in period t at site s .
- $\text{H2_pipe_new_capacity}(t,s)$ – capacity of H2 pipe built in period t at site s .
- $\text{H2_pipe_available_capacity}(t,s)$ – available capacity of H2 pipe in period t at site s .
- $\text{CO2_pipe_new_capacity}(t,s)$ – capacity of CO2 pipe built in period t at site s .
- $\text{CO2_pipe_available_capacity}(t,s)$ – available capacity of CO2 pipe in period t at site s .

Exogenous variables

- $\text{residual_capacity}(t,s,\text{tech})$ – capacity of technology tech , in period t , at site s , which exists in the base (start) year. Also referred to as ‘starting’ or ‘existing’ capacity.
- $\text{lifetime}(\text{tech})$ – the length of time in periods t , after which a unit of capacity built expires.
- $\text{demand}(t,s)$ – the required demand/capacity of the end process at site s in period t .
- $\text{availability_factor}(\text{tech})$ – the amount of available technology tech which can actually be used.

Constraints

The essential model constraints that govern how the model works are outlined here (optional user constraints are not covered). Their implementation in the model is explained in more detail in the next section.

- **Capacity transfer constraint:**
 - $\text{available_capacity}(t,s,\text{tech}) = (\sum \text{new_capacity}(t',s,\text{tech}))$ for all periods t' preceding t by the $\text{lifetime}(\text{tech}) + \text{residual_capacity}(t,s,\text{tech})$
- **Availability constraint:**
 - $\text{used_capacity}(t,s,\text{tech}) < \text{available_capacity}(t,s,\text{tech}) * \text{availability_factor}(\text{tech}) * \text{capacity_to_activity_factor}(\text{tech})$
- **Production constraints:**

Production technologies at a site can be grouped into processes which produce the same output commodity. We denote processes which produce required end demand commodity ‘ d ’, process_d . The first production constraint ensures the site produces the correct amount of end demand commodity ‘ d ’ (e.g. cement). The second production constraint ensures that each process at a site produces the correct amount of commodity for inputs to other processes. This constraint requires that the output commodity produced by any process at a site (other than the process producing the end demand commodity) is equal to the amount of that commodity that is used by other technologies at a site.

 - $\sum \text{used_capacity}(t,s,\text{tech}) = \text{demand}(t,s)$
for $\text{tech} \in \text{process}_d(s)$
 - $\sum \text{used_capacity}(t,s,\text{tech}_a) = \sum (\text{used_capacity}(t,s,\text{tech}_b) * \text{input_amount}(\text{tech}_b))$, for any output commodity, for $\text{tech}_a \in \text{process}(s)$ which is not $\text{process}_d(s)$
- **H2 pipe constraints:**
 - $\text{H2_pipe_available_capacity}(t,s) = (\sum \text{H2_pipe_new_capacity}(t',s))$ for all periods t' preceding t by the $\text{lifetime}(\text{H2 pipe})$
 - $\text{H2_pipe_available_capacity}(t,s) \geq \sum \text{used_capacity}(t,s,\text{tech}) * \text{hydrogen_fuel_share}(\text{tech})$

- **CO2 pipe constraints:**
 - $\text{CO2_pipe_available_capacity}(t,s) = (\sum \text{CO2_pipe_new_capacity}(t,s))$ for all periods t preceding t by the lifetime(CO2 pipe)
 - $\text{CO2_pipe_available_capacity}(t,s) \geq \sum \text{used_capacity}(t,s,\text{tech}) * \text{captured_emissions}(\text{tech})$

Variable bounds

- No negative capacity
 - $\text{new_capacity}(t,s,\text{tech}) \geq 0$
 - $\text{available_capacity}(t,s,\text{tech}) \geq 0$
 - $\text{used_capacity}(t,s,\text{tech}) \geq 0$
 - $\text{H2_pipe_available_capacity}(t,s) \geq 0$
 - $\text{H2_pipe_new_capacity}(t,s) \geq 0$
 - $\text{CO2_pipe_available_capacity}(t,s) \geq 0$
 - $\text{CO2_pipe_new_capacity}(t,s) \geq 0$

As well as the site-level variables and constraints, there are also two national-level components: (i) CO2 transport and storage from cluster to storage site, and (ii) national hydrogen production and transportation.

National CO2 transport and storage

Some CO2 produced at a cluster can be transported and stored at a storage site. The per unit cost of storing CO2 is constant but varies by storage location.

- **CO2_transported($t,c,\text{term},\text{store}$)** – the amount of CO2 transported from cluster c to storage site store via terminal term in period t .

Exogenous variables

- $\text{CO2_transport_cost}(c,\text{term},\text{store})$ – the cost of transporting one unit of CO2 from cluster c to storage site store via terminal term .
- $\text{CO2_non_industry}(t,c)$ – the amount of captured carbon from the non-industry sector at time t at cluster c .

Constraints

- Storage site injection constraint
 - $\sum \text{CO2_transported}(t,c,\text{term},\text{store})$ for each store $\leq \text{max_injection}(t,\text{store})$
- Site-cluster relationship
 - $\sum \text{CO2_transported}(t,c,\text{term},\text{store})$ for each cluster $c = \sum \text{CO2_captured}(t,s)$ for all sites s in cluster $c + \text{CO2_non_industry}(t,c)$.

Variable bounds

- Positive injection only
 - $\text{CO2_transported}(t,c,\text{term},\text{store}) \geq 0$

National hydrogen production and transport modelling

COMIT includes the capability to explicitly model hydrogen production. In this 'hydrogen production mode', hydrogen is produced at plants located at the centre of each cluster. Additionally, pipes can be built between clusters to transport hydrogen from one cluster to another. At each cluster, a hydrogen plant is then modelled as a site as above.

However, the hydrogen production component of the model changes the model to a mixed integer linear program (a MILP, rather than just an LP) and, as a result, greatly increases solution time. It should be used sparingly. In most cases, COMIT is run with this functionality turned off via the `model_H2_production` parameter (set to false) in the **model_parameters_b** tab.

Objective function

The objective of the model is to maximise national NPV over the modelling time horizon while meeting industrial demand (e.g. the amount of cement required) at each site. Since sites do not receive revenue, we can minimise the present value of all costs, under the assumption that costs are positive numbers. This is done by minimising the present value (PV) of all costs, summed across all t timesteps in the time horizon (e.g. 2021–2050).

$$\min_{t \in \mathbb{N}} \sum_{t=1}^{t_{max}} PV(t)$$

where $t=1$ is the initial year (i.e. 2021), t_{max} is the final time period (i.e. 2050), and N is the number of timesteps in the model.

The PV to be minimised is the sum of the 8 different cost types:

$$PV(t) = PV_{\text{technology capex}}(t) + PV_{\text{fixed opex}}(t) + PV_{\text{fuel costs}}(t) + PV_{\text{carbon cost}}(t) + PV_{\text{site H2 T\&S}}(t) \\ + PV_{\text{site CO2 T\&S}}(t) + PV_{\text{national H2 T\&S}}(t) + PV_{\text{national CO2 T\&S}}(t)$$

- **$PV_{\text{technology capex}}$** : the cost of buying a unit of new technological capacity
- **$PV_{\text{fixed opex}}$** : the operating cost of having a unit of technological capacity available on site
- **$PV_{\text{fuel costs}}$** : the cost of the fuel needed to use a unit of technological capacity
- **$PV_{\text{carbon cost}}$** : the emissions emitted at each site multiplied by a carbon value
- **$PV_{\text{site H2 T\&S}}$** : the combined capex and opex cost of a unit of hydrogen pipe built between a cluster and a site, to transport hydrogen from a site to cluster point
- **$PV_{\text{site CO2 T\&S}}$** : the combined capex and opex cost of a unit of CO2 pipe built between a site and cluster
- **$PV_{\text{national H2 T\&S}}$** : the combined capex and opex cost of a unit of H2 pipe built between two clusters
- **$PV_{\text{national CO2 T\&S}}$** : the cost of transport and storage for a unit of CO2 moved from cluster to a storage site

The first six terms are site-level, and the last two are system-wide, summed across all unique H2/CO2 routes from cluster to terminal to storage site.

More information about how each of these terms is calculated is provided below.

PV_technology_capex

This is the amount of new capacity built in a year, multiplied by the capex (capital) cost of the technology. It is assumed sites borrow money to pay for the capital expenditure. Thus, for a given period t , the site is paying for investments made in a period $t-n$ if the current period falls into the *term* of the loan.

$$PV_{\text{technology capex}}(t) = \sum_s \sum_{tech} \sum_{t=t_{tech}}^{t-term_{tech}} \text{annual_payment}_t(\text{initial principal}, \text{interest}, \text{term})$$

Expressing annual payment using an interest rate r :

$$PV_{technology\ capex}(t) = \sum_s \sum_{tech} \sum_t^{t-term_{tech}} initial_principal * \frac{r(1+r)^{term_{tech}}}{(1+r)^{term_{tech}} - 1}$$

The initial principal can be expressed as new capacity. Capex costs can be discounted to the base year.

$$PV_{technology\ capex}(t) = \sum_s \sum_{tech} \sum_t^{t-term_{tech}} new_capacity(t, s, tech) * capex_t(tech) * \frac{r(1+r)^{term_{tech}}}{(1+r)^{term_{tech}} - 1}$$

Technology capital costs (per unit of installed capacity) are listed in the Technologies tab of the Data input workbook.

PV_fixed_opex

Fixed opex (i.e. operating cost) is an annual cost that must be paid for any installed capacity (available capacity) at each site. These costs are specified for each technology in the fixed_opex column in the Technologies sheet of the Data input worksheet. Note that it is paid on all available technological capacity, regardless of how much it is used that year.

It can be expressed as:

$$PV_{fixed\ opex}(t) = \sum_{tech} fixed_opex(tech, t) * \sum_s available_capacity(t, s, tech)$$

Let $fixed_opex_t$ be the discounted cost in period t and $fixed_opex_0$ be the cost in the base year:

$$PV_{fixed\ opex}(t) = \sum_{tech} \frac{fixed_opex_0(tech)}{(1 + discount_rate)^t} * \sum_s available_capacity(t, s, tech)$$

PV_fuel_cost

This is equal to the amount of fuel used by each technology at each site, multiplied by the fuel costs.

$$PV_{fuel\ costs}(t) = \sum_s \sum_{tech} \sum_{fuel} used_capacity(t, s, tech) * fuel_use(fuel, tech) * fuel_cost_t(fuel)$$

Fuel costs are specified for each year and fuel in the fuel_costs tab of the Data input workbook. Sites can use technologies which generate electricity. A certain amount of this electricity is used within the site, the rest is sold to the grid. If you assume the price of electricity is the same for buying and selling, it does not matter how much is used on site, just assume everything is sold to grid.

$$PV_{fuel\ costs}(t) = \sum_s \sum_{gen_tech} used_capacity(t, s, tech) * generation(tech) * -fuel_cost_t(electricity, t) + \sum_{tech} \sum_{fuel} used_capacity(t, s, tech) * fuel_use(fuel, tech) * fuel_cost_t(fuel, t)$$

Where $gen_tech \in$ technologies which can generate electricity.

When discounting fuel costs, consider that some fuels can change price annually.

PV_carbon_cost

This is the amount of carbon emitted at each site multiplied by an annual carbon value. A carbon value is included in the NPV calculation to represent a trade-off between costs and emissions.

$$PV_{\text{carbon cost}}(t) = \text{carbon_price}(t) * \sum_s \text{released_emissions}(t, s)$$

Carbon emitted at each site can be broken down into process and fuel emissions.

$$PV_{\text{carbon cost}}(t) = \text{carbon_price}(t) * \sum_s \text{released_process_emissions}(t, s) + \text{released_fuel_emissions}(t, s)$$

$$PV_{\text{carbon cost}}(t) = \text{carbon_price}(t) * \left(\sum_s \sum_{\text{tech}} \text{used_capacity}(t, s, \text{tech}) * \text{process_emissions}(\text{tech}) * (1 - \text{capture_rate}(\text{tech})) + \sum_s \sum_{\text{tech}} \sum_{\text{fuel}} \text{used_capacity}(t, s, \text{tech}) * \text{fuel_use}(\text{fuel}, \text{tech}) * \text{fuel_emissions}(\text{fuel}) * (1 - \text{capture_rate}(\text{tech})) \right)$$

Fuel emissions are specified in the **Fuel_emissions** tab in the Data input workbook, in terms of greenhouse gas emissions (kt) per unit of commodity consumed (PJ). As with fuel costs, for some fuels, this rate changes annually.

PV_site_H2_T&S

This term represents a site's H2 transport and storage costs. Hydrogen pipes between a cluster and site are modelled in much the same way as an industrial technology. There is a capital cost and a fixed operating cost, which is paid on the total installed capacity.

$$PV_{\text{site H2 T\&S}}(t) = \sum_s \text{H2_pipe_opex}(t, s) + \text{H2_pipe_capex}(t, s)$$

The amount of new H2 pipe built in a year is then multiplied by the capex cost of the pipe. It is assumed sites borrow money to pay for the capital expenditure over the lifetime of the pipe. Thus, for a given period t , the site is paying for investments made in a period $t-n$ if the current period falls into the *term* of the loan.

$$PV_{\text{site H2 T\&S}}(t) = \sum_s \text{H2_pipe_opex}(t, s) + \sum_t \text{H2_pipe_new_capacity}(t, s) * \text{H2_pipe_capex}_t(s, \text{capacity}) * \frac{r(1+r)^{\text{term}_{\text{H2 pipe}}}}{(1+r)^{\text{term}_{\text{H2 pipe}}} - 1}$$

Opex must be paid on the entire installed capacity (i.e. available capacity) of the pipe, regardless of whether it is used in that timestep.

$$PV_{site\ H2\ T\&S}(t) = \sum_s \sum_{t-term_{H2\ pipe}}^{t-1} H2_pipe_available_capacity(t, s) * H2_pipe_opex_t(s, capacity) \\ + \sum_t H2_pipe_new_capacity(t, s) * H2_pipe_capex_t(s, capacity) \\ * \frac{r(1+r)^{term_{H2\ pipe}}}{(1+r)^{term_{H2\ pipe}} - 1}$$

H2 pipe costs per unit of installed pipe are included in the **H2_transport_cost** tab of the Data input workbook. While not included above, to allow this section to focus on the key concepts governing the model, more recent iterations of the model have also included the option to use trucking and the national grid system to transport H2. How the choice of transportation method and corresponding costs are calculated is explained in the section on H2 transport.

PV_site_CO2_T&S

This term represents a site's CO2 transportation cost from site to cluster centre. We can model CO2 transport pipes like hydrogen pipes.

$$PV_{site\ CO2\ T\&S}(t) = \sum_s CO2_pipe_capex(t, s) + CO2_pipe_fixed_opex(t, s)$$

Start by modelling pipe capex:

$$PV_{site\ CO2\ T\&S}(t) = \sum_s \sum_t^{t-term_{CO2\ pipe}} \left(CO2_pipe_new_capacity(t, s) * CO2_pipe_capex_t(s, capacity) \right. \\ \left. * \frac{r(1+r)^{term_{H2\ pipe}}}{(1+r)^{term_{H2\ pipe}} - 1} \right) + CO2_pipe_fixed_opex(t, s)$$

Fixed opex is paid on the entire installed capacity:

$$PV_{site\ CO2\ T\&S}(t) = \sum_s \sum_t^{t-term_{CO2\ pipe}} \left(CO2_pipe_new_capacity(t, s) * CO2_pipe_capex_t(s, capacity) \right. \\ \left. * \frac{r(1+r)^{term_{H2\ pipe}}}{(1+r)^{term_{H2\ pipe}} - 1} \right) \\ + (CO2_pipe_available_capacity(t, s) * CO2_pipe_fixed_opex_t(s, capacity))$$

The CO2 capex and opex costs per unit of installed pipe are input in the **CO2_transport_cost** tab of the Data input workbook. As with H2 transportation, options for CO2 trucking have subsequently been added to the model and are detailed later in this guide.

PV_national_CO2_T&S

This term covers the cost for the transportation of CO2 from cluster to storage site. The cost of transporting CO2 from the cluster to the storage site is the product of the amount of CO2 transported via each route, multiplied by the per unit cost of the route.

$$PV_{national\ CO2\ T\&S}(t) = \sum_r CO2_transported(t, c, term, store) * CO2_transport_cost_t(c, term, store)$$

Where r is a unique CO2 route from cluster to terminal to storage site. These costs (£m/kt) for transport and storage of CO2 are set out for all possible shipping, transport and storage routes in the **CO2_T&S_cost** tab of the Data input workbook.

Constraints

Essential constraints

Capacity transfer constraint

As used in COMIT, ‘capacity’ refers to the amount of output of a commodity that an industrial technology or technological process (e.g. a coal power plant) can produce. Typical units are PJ/year for most energy technologies and million tonnes per year for steel, aluminium and paper industries.

The capacity transfer constraint governs the amount of technological capacity available for use at each timestep, site and technology throughout the model run.

$\text{available_capacity}(t,s,\text{tech}) = (\sum \text{new_capacity}(t',s,\text{tech}))$ for all periods t' preceding t by the $\text{lifetime}(\text{tech}) + \text{residual_capacity}(t,s,\text{tech})$

The above equation states that the amount of technological capacity available at site s in timestep t is equal to the sum of all investments in new capacity up to that period, where the physical life has not yet ended and the capacity is still in operating condition, plus any residual capacity that is still available. The residual capacity is the amount of existing capacity for each technology and site already present in the model at the starting year (e.g. 2021).

The capacity transfer constraint is thus the way in which the model calculates the amount of technological capacity available (i.e. $\text{available_capacity}(t,s,\text{tech})$) at each site in each year.

In reality, new technological capacity would be built in discrete blocks, such as installing a new boiler or heat pump at a site. However, in COMIT, capacities are assumed to be built incrementally rather than all at once.

Once built, any existing capacity is assumed to last its full lifetime at full capacity unless replaced.

The exception to this is the residual capacity present at the model start year, which is assumed to reduce linearly over the course of its lifetime. In reality, the residual technological capacities across industrial sites are likely to be at a range of points in their lifetime. However, as no information is available about this existing capacity distribution, no specific remaining lifetime is allocated to individual sites and instead COMIT assumes the residual capacities all decrease linearly.

$\text{residual_capacity}(t,s,\text{tech}) = \max(0, [-\text{residual_capacity}(2021,s,\text{tech}) / \text{lifetime}(\text{tech})] * (t-2021))$

Note that the capacity transfer constraint also applies to Hydrogen and CO2 pipe capacity (though for pipes, there is no pipe residual capacity at the model start year):

$\text{H2_pipe_available_capacity}(t,s) = (\sum \text{H2_pipe_new_capacity}(t',s,\text{tech}))$ for all periods t' preceding t by the $\text{lifetime}(\text{tech})$

$\text{CO2_pipe_available_capacity}(t,s) = (\sum \text{CO2_pipe_new_capacity}(t',s,\text{tech}))$ for all periods t' preceding t by the $\text{lifetime}(\text{tech})$

Availability constraint

This constrains the activity (i.e. used capacity) for each technology, in year t , at site s , so that it does not exceed the available technological capacity:

$$\text{used_capacity}(t,s,\text{tech}) \leq \text{available_capacity}(t,s,\text{tech}) * \text{availability_factor}(\text{tech}) * \text{capacity_to_activity_factor}(\text{tech})$$

This means that in each timestep the model may use some or all the installed technological capacity at each site according to the Availability Factor (AF) of that technology. Note that the model may decide to use less than the available capacity during certain timesteps, if such a decision contributes to minimising the overall cost.

Example: a coal-fired power plant's activity in any timestep is at most 80% of its available capacity, i.e. it has an availability_factor of 0.8 and thus

$$\text{used_capacity}(t,s,\text{tech}) \leq 0.8 * \text{capacity_to_activity_factor}(\text{tech}) * \text{available_capacity}(t,s,\text{tech})$$

Production constraints

These constrain the production rate of a process technology so that production by each site equals the amount consumed at the site (i.e. the quantity produced is equal to the quantity consumed).

Technologies at a site can be grouped into processes which produce the same output commodity. Let us call the processes which produce a particular end demand commodity, process_d .

The first production constraint says that at each site, in each year, the amount of end demand commodity produced must be equal to the exogenous demand (i.e. $\text{demand}(t,s)^3$).

$$\sum \text{used_capacity}(t,s,\text{tech}) = \text{demand}(t,s) \text{ for } \text{tech} \in \text{process}_d(s)$$

The second production constraint ensures that each process at a site produces the correct amount of commodity for inputs to other processes. The second constraint requires that the output commodity produced by any process at a site (other than the process producing the end demand commodity) is equal to the amount of that commodity used by other technologies at a site.

$$\sum \text{used_capacity}(t,s,\text{tech}_a) = \sum (\text{used_capacity}(t,s,\text{tech}_b) * \text{input_amount}(\text{tech}_b)), \text{ for any output commodity, for } \text{tech}_a \in \text{process}(s) \text{ which is not } \text{process}_d(s)$$

H2_site_cluster_availability constraint

This constraint states that the available pipe capacity to transport hydrogen for site s in timestep t must be greater than or equal to the amount of hydrogen produced at that site and timestep. It ensures that there is enough hydrogen pipe capacity to transport all the hydrogen produced.

$$\text{H2_pipe_available_capacity}(t,s) \geq \sum (\text{used_capacity}(t,s,\text{tech}) * \text{hydrogen_fuel_share}(\text{tech}))$$

The above constraint applies to those sites where the cheapest option for transporting hydrogen is to build pipelines (by default, building pipelines is set to be the cheapest option for all sites).

For sites where trucking or repurposed gridlines are the cheapest option, the constraint becomes either:

$$\text{H2_truck_available_capacity}(t,s) \geq \sum (\text{used_capacity}(t,s,\text{tech}) * \text{hydrogen_fuel_share}(\text{tech}))$$

$$\text{H2_grid_available_capacity}(t,s) \geq \sum (\text{used_capacity}(t,s,\text{tech}) * \text{hydrogen_fuel_share}(\text{tech}))$$

³ See site demand section for more detail

CO2_availability

This constraint ensures that each site has enough CO2 transport infrastructure (pipes or trucking) to handle the amount of CO2 captured by CCS technologies at that site.

$$\text{CO2_pipe_available_capacity}(t,s) \geq \sum (\text{technology_used_capacity}(t,s,\text{tech}) * \text{captured_emissions}(\text{tech}))$$

The above constraint applies to sites where the pre-specified option for transporting CO2 is to build pipelines. For sites where trucking is the pre-specified option, the constraint becomes:

$$\text{CO2_truck_available_capacity}(t,s) \geq \sum (\text{technology_used_capacity}(t,s,\text{tech}) * \text{captured_emissions}(\text{tech}))$$

CO2_cluster_storage_site_relationship

This constraint states that the amount of CO2 transported from cluster c to storage sites in period t must equal the total amount of CO2 captured across all sites in that cluster, after accounting for non-industry CO2 demand. It ensures that all CO2 captured in a cluster is transported to storage or a terminal.

$$\sum \text{CO2_transported}(t,c,\text{term},\text{store}) = \sum \text{CO2_captured}(t,s) + \text{CO2_non_industry}(t,c).$$

where $\sum \text{CO2_transported}(t,c,\text{term},\text{store})$ is the sum of CO2 transported across all storage sites store and terminals term in period t for cluster c .

Storage site injection constraint

This constraint ensures that the annual capacity of CO2 that can be injected into each storage site is not exceeded. This reflects physical and regulatory limits on storage capacity.

$$\sum \text{CO2_transported}(t,c,\text{term},\text{store}) \leq \text{max_injection}(t,\text{store})$$